

HR Analytics Project: Employee Attrition Prediction and Prevention

1. Project Overview

Employee attrition is a key concern for HR departments. High attrition rates can lead to increased recruitment costs, loss of experienced employees, and reduced organizational efficiency. This project aims to build a predictive model to identify employees likely to leave the organization and provide actionable insights to help reduce attrition.

2. Data Preprocessing & Feature Engineering

- Dataset loaded from HR records (CSV format).
 - Removed redundant features (e.g., EmployeeCount, Over18, StandardHours).
 - Encoded categorical variables using one-hot encoding.
 - Created new features where relevant (e.g., total working years, promotion history).
 - Balanced the target variable (Attrition) using stratified train-test split.
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3. Model Building

Target Variable: Attrition (Yes = 1, No = 0)

Features: All columns excluding Attrition

Train-Test Split: 80% training, 20% testing, stratified on the target.

Models Evaluated:

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.864	0.845	0.864	0.844
Decision Tree	0.820	0.782	0.820	0.795
Random Forest	0.844	0.806	0.844	0.791

Best Performing Model: Logistic Regression

4. Confusion Matrices

Visualized confusion matrices for each model to evaluate prediction balance and errors:

- Logistic Regression: Best balance between precision and recall.
 - Decision Tree: Slightly more false negatives.
 - Random Forest: High recall but slightly lower precision.
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5. Model Explainability with SHAP

Tool Used: SHAP (SHapley Additive exPlanations)

- Computed SHAP values using the best model (Logistic Regression).
- Generated a summary plot to rank feature importance.

Top Predictive Features:

- OverTime
- JobSatisfaction
- EnvironmentSatisfaction
- YearsSinceLastPromotion
- MonthlyIncome

Interpretation:

- Employees with high overtime and low satisfaction levels are more likely to leave.
 - Long gaps in promotions and lower income contribute to attrition.
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6. Power BI Dashboard

Data Source: Preprocessed dataset (CSV format)

Visualizations Included:

- Attrition rate by department and job role (bar chart)
- SHAP-based top features (bar chart)
- Interactive filters (slicers) for gender, department, job role, etc.

Purpose:

To allow HR teams to explore attrition patterns and risks interactively.

7. Deliverables

1. Power BI Dashboard

- Fully interactive
- Real-time filtering and visual insights

2. Model Evaluation Report (PDF)

- Model performance comparison
- Confusion matrices and feature importance plots
- Attrition Prevention Suggestions **Insights:** (Overtime, satisfaction, and promotion delays are key attrition drivers).

Recommendations:

- Monitor overtime patterns
 - Launch satisfaction improvement programs
 - Offer clear paths for promotions and recognition
 - Use predictions for proactive HR interventions
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Conclusion

This project demonstrates the power of predictive analytics in HR decision-making. By identifying key attrition drivers and high-risk employees, organizations can implement data-driven strategies to improve retention and employee satisfaction.

Tools Used:

- Python (Pandas, Sklearn, SHAP)
- Power BI
- Matplotlib & Seaborn (for visuals)

Outcome: A full ML pipeline with actionable insights for HR leadership.